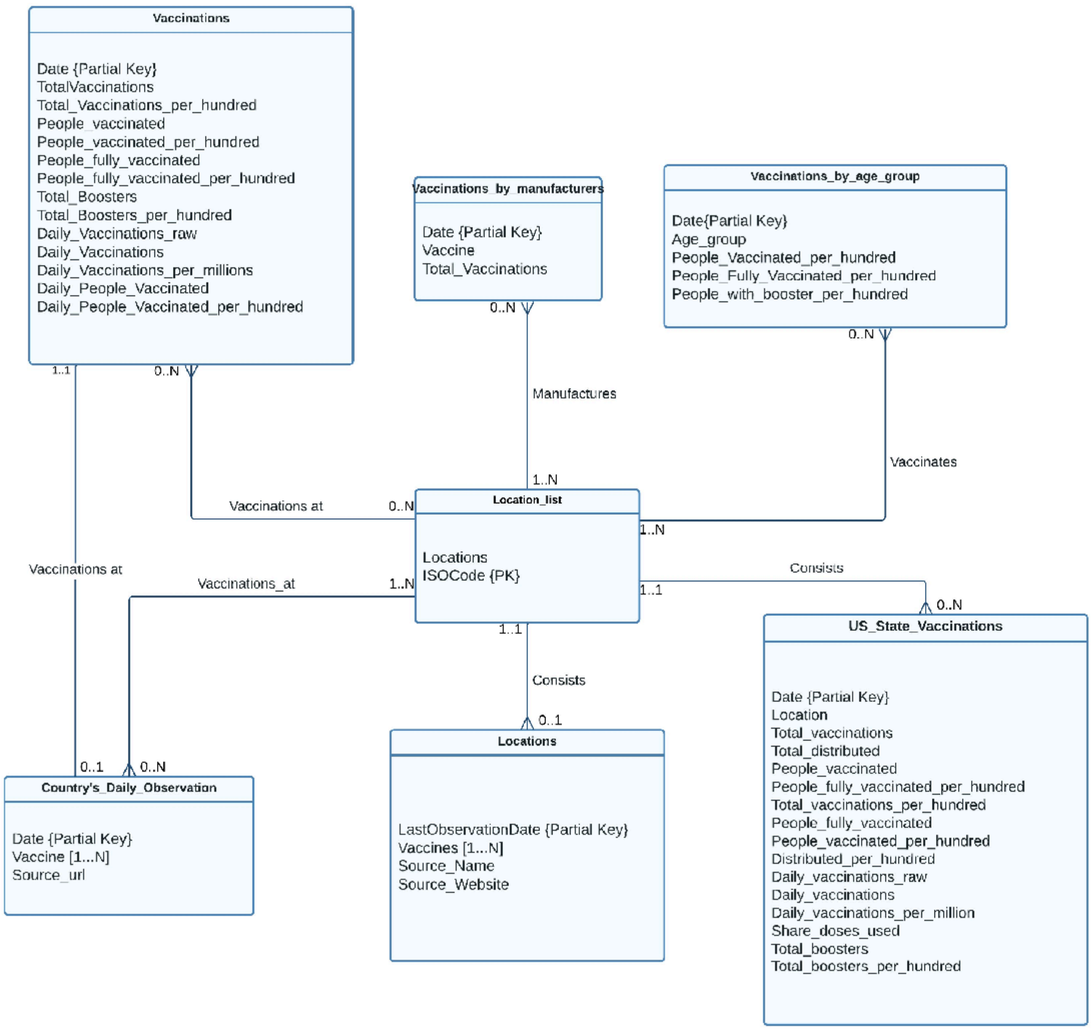
1: Designing an Entity Relationship Model

Entity Relationship Model for Data on COVID-19 (coronavirus) vaccinations by Our World



Assumptions:

- As given in the requirement, we need to have an entity of four countries, Australia, Germany, Italy, and United States, but we have the same details in the Vaccinations Entity
- I have created an entity called Country's_Daily_observation and have combined the data of Australia, Germany, Italy and United States into single table and rest of the information can be fetched from Vaccinations Entity
- I have created a new entity called Location_list with Locations and ISO_Code as attributes where ISO_Code will be the primary key of the entity

- Instead of having the location names in the entities, I have planned to replace them with ISO_Code
- Here, the count of location name in Locations entity and the count of unique location names in Vaccinations does not equal, So I have taken the ISO_Code from Vaccinations entity as well to create the Location_list entity
- We have multi-valued attributes in some entities will be managed in later part of the Schema

2: Mapping an ER Model to a Relational Database Schema

<u>Step 1: Strong Entities</u>

- Location_list (<u>ISO_Code</u>, Locations)

<u>Step 2: Weak Entities</u>

- Locations (<u>ISO_Code*, LastObservationDate</u>, Vaccines [1...N], Source_Name, Source_Website)
- Vaccinations (<u>ISO_Code*, Date</u>, TotalVaccinations, Total_Vaccinations_per_hundred, People_vaccinated, People_vaccinated_per_hundred, People_fully_vaccinated, People_fully_vaccinated_per_hundred, Total_Boosters, Total_Boosters_per_hundred, Daily_Vaccinations_raw, Daily_Vaccinations, Daily_Vaccinations_per_millions, Daily_People_Vaccinated, Daily_People_Vaccinated_per_hundred)
- Vaccinations_by_manufacturers (<u>ISO_Code*, Date, Vaccines</u>, Total_Vaccinations)
- Vaccinations_by_age_group (<u>ISO_Code*, Date, Age_group</u>, People_Vaccinated_per_hundred, People_Fully_Vaccinated_per_hundred, People_with_booster_per_hundred)
- US_State_Vaccinations (<u>ISO_Code*,Date, Location</u>, Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred)
- Country's_Daily_Observation (<u>ISO_Code*, Date, </u>Vaccine [1...N], Source_url)

Step 3: One-to-One Relationships

Locations has a 1…1 relationship with Location_list, but ISO_Code is already included in Locations as a foreign key, together with LastObservationDate makes the primary key of Locations.

- Locations (ISO_Code*, LastObservationDate, Vaccines [1...N], Source_Name, Source_Website)

US_State_Vaccinations has a 1…1 relationship with Location_list, but ISO_Code is already included in US_State_Vaccinations as a foreign key, together with Date, Location makes the primary key of US_State_Vaccinations.

- US_State_Vaccinations (ISO_Code*,Date, Location, Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred)

Step 4: One-to-Many Relationships

As we have already included ISO_Code as foreign key with 1...N relationship, so no further action is required.

Step 5: Many-to-Many Relationships

Nothing to do here, we don't have any Many to Many relationship

Step 6: Multi-valued Attributes

As we can see in Locations, Vaccinations_by_age_group and Country's_Daily_Observation have Multi-valued Attributes of Vaccines

So, I have created a new entity named Vaccine_list, which consist of all the unique types of vaccines which are given in above listed four entities and have created a Vaccine_ID for all the individual Vaccines.

- Vaccine_list (Vaccine_ID, Vaccine_Name)

Step 7: Higher Degree Relationships

Nothing to do here, we do not have any ternary relationship

Special Case:

Nothing to do here, we do not have any Special Cases.

Relational Database Schema:

- Location_list (ISO_Code, Locations)
- Locations (ISO_Code*, LastObservationDate, Vaccines [1...N], Source_Name, Source_Website)
- Vaccinations (ISO_Code*, Date, TotalVaccinations, Total_Vaccinations_per_hundred, People_vaccinated, People_vaccinated_per_hundred, People_fully_vaccinated, People_fully_vaccinated_per_hundred, Total_Boosters, Total_Boosters_per_hundred, Daily_Vaccinations_raw, Daily_Vaccinations, Daily_Vaccinations_per_millions, Daily_People_Vaccinated, Daily_People_Vaccinated_per_hundred)
- Vaccinations_by_manufacturers (ISO_Code*, Date, Vaccine, Total_Vaccinations)
- Vaccinations_by_age_group (ISO_Code*, Date, Age_group, People_Vaccinated_per_hundred, People_Fully_Vaccinated_per_hundred, People_with_booster_per_hundred)
- US_State_Vaccinations (ISO_Code*,Date, Location, Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred)
- Country's_Daily_Observation (ISO_Code*, Date, Vaccine [1...N], Source_url)
- Vaccine_list (Vaccine_ID, Vaccine_Name)

Normalisation:

1. Functional Dependencies:
   - ISO_Code -> Locations
   - ISO_Code, LastObservationDate -> Vaccines [1...N], Source_Name, Source_Website
   - ISO_Code, Date -> TotalVaccinations, Total_Vaccinations_per_hundred, People_vaccinated, People_vaccinated_per_hundred, People_fully_vaccinated, People_fully_vaccinated_per_hundred, Total_Boosters, Total_Boosters_per_hundred, Daily_Vaccinations_raw, Daily_Vaccinations, Daily_Vaccinations_per_millions, Daily_People_Vaccinated, Daily_People_Vaccinated_per_hundred
   - ISO_Code, Date, Vaccines -> Total_Vaccinations

- ISO_Code, Date, Age_group -> People_Vaccinated_per_hundred, People_Fully_Vaccinated_per_hundred, People_with_booster_per_hundred
- ISO_Code, Date, Location -> Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred
- ISO_Code, Date -> Vaccine [1...N], Source_url
- Vaccine_ID -> Vaccine_Name

2. Highest Normal Form:

- Location_list (ISO_Code, Locations) -> 3NF
- Locations (ISO_Code*, LastObservationDate, Vaccines [1...N], Source_Name, Source_Website)

  -- A Non primary key attribute has a multi-valued attribute, failed 1NF

- Vaccinations (ISO_Code*, Date, TotalVaccinations, Total_Vaccinations_per_hundred, People_vaccinated, People_vaccinated_per_hundred, People_fully_vaccinated, People_fully_vaccinated_per_hundred, Total_Boosters, Total_Boosters_per_hundred, Daily_Vaccinations_raw, Daily_Vaccinations, Daily_Vaccinations_per_millions, Daily_People_Vaccinated, Daily_People_Vaccinated_per_hundred) -> 3NF
- Vaccinations_by_manufacturers (ISO_Code*, Date, Vaccine, Total_Vaccinations) -> 3NF
- Vaccinations_by_age_group (ISO_Code*, Date, Age_group, People_Vaccinated_per_hundred, People_Fully_Vaccinated_per_hundred, People_with_booster_per_hundred) -> 3NF
- US_State_Vaccinations (ISO_Code*,Date, Location, Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred) -> 3NF
- Country's_Daily_Observation (ISO_Code*, Date, Vaccine [1...N], Source_url)

  -- A Non primary key attribute has a multi-valued attribute, failed 1NF

- Vaccine_list (Vaccine_ID, Vaccine_Name) -> 3NF

3. Decomposing into 3NF relations

   Using Text to Columns option in Microsoft Excel, I have separated multi-valued attributes into columns.

   After splitting the comma separated values in Vaccine column, I used Pivot-Longer function to change to columns into row, so all the individual vaccines used will be in each tuple for the respective location and date

- Location_list (<u>ISO_Code</u>, Locations) -> 3NF
- Locations (<u>ISO_Code*, LastObservationDate, Vaccines</u>, Source_Name, Source_Website) -> 3NF
- Vaccinations (<u>ISO_Code*, Date</u>, TotalVaccinations, Total_Vaccinations_per_hundred, People_vaccinated, People_vaccinated_per_hundred, People_fully_vaccinated, People_fully_vaccinated_per_hundred, Total_Boosters, Total_Boosters_per_hundred, Daily_Vaccinations_raw, Daily_Vaccinations, Daily_Vaccinations_per_millions, Daily_People_Vaccinated, Daily_People_Vaccinated_per_hundred) -> 3NF
- Vaccinations_by_manufacturers (<u>ISO_Code*, Date, Vaccine</u>, Total_Vaccinations) -> 3NF
- Vaccinations_by_age_group (<u>ISO_Code*, Date, Age_group</u>, People_Vaccinated_per_hundred, People_Fully_Vaccinated_per_hundred, People_with_booster_per_hundred) -> 3NF
- US_State_Vaccinations (<u>ISO_Code*,Date, Location</u>, Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred) -> 3NF
- Country's_Daily_Observation (<u>ISO_Code*, Date, Vaccine,</u> Source_url)
- Vaccine_list (<u>Vaccine_ID</u>, Vaccine_Name) -> 3NF

4. Final Relationship Schema:

   I have created a relation between the entities which has the vaccine names with the Vaccine_List entity and replaced all the vaccine names with Vaccine_ID and link the relation

   Below given is the Final Relationship Schema:

   Location_list (<u>ISO_Code</u>, Locations)

   Vaccine_list (<u>Vaccine_ID</u>, Vaccine_Name)

   Locations (<u>ISO_Code, LastObservationDate, Vaccine_ID,</u> Source_Name, Source_Website)*

   Vaccinations (<u>ISO_Code, Date</u>, TotalVaccinations, Total_Vaccinations_per_hundred, People_vaccinated, People_vaccinated_per_hundred, People_fully_vaccinated, People_fully_vaccinated_per_hundred, Total_Boosters, Total_Boosters_per_hundred, Daily_Vaccinations_raw, Daily_Vaccinations, Daily_Vaccinations_per_millions, Daily_People_Vaccinated, Daily_People_Vaccinated_per_hundred)*

   Vaccinations_by_manufacturers (<u>ISO_Code, Date, Vaccine_ID</u>, Total_Vaccinations)*

   Vaccinations_by_age_group (<u>ISO_Code, Date, Age_group</u>, People_Vaccinated_per_hundred, People_Fully_Vaccinated_per_hundred, People_with_booster_per_hundred)*

   US_State_Vaccinations (<u>ISO_Code,Date, Location</u>, Total_vaccinations, Total_distributed, People_vaccinated, People_fully_vaccinated_per_hundred, Total_vaccinations_per_hundred, People_fully_vaccinated, People_vaccinated_per_hundred, Distributed_per_hundred, Daily_vaccinations_raw, Daily_vaccinations, Daily_vaccinations_per_million, Share_doses_used, Total_boosters, Total_boosters_per_hundred)*

   Country's_Daily_Observation (<u>ISO_Code, Date, Vaccine_ID</u>, Source_url)*